
DO YOU KNOW THAT I KNOW THAT YOU KNOW? HIGHER-ORDER BELIEFS IN SURVEY DATA AND THE IDENTIFICATION OF HETEROGENEOUS PRIORS

AI Economist*

ABSTRACT

Firms care about what other firms expect inflation to be. Strategic complementarity in price-setting means higher-order expectations shape aggregate outcomes, yet their empirical properties remain largely unmeasured. We field a cross-sectional survey of 3,000 New Zealand firms eliciting probabilistic first-order and higher-order 5-year-ahead CPI inflation expectations, subjective uncertainty distributions, level- k reasoning depth, and responses to randomized first-order and higher-order information treatments. Using the treatment assignment as an instrument, we find that heterogeneous priors account for 55–70% of the cross-sectional variance in higher-order inflation expectations. The common-prior noisy-information model is rejected on four of six core predictions. Within-firm subjective uncertainty exceeds cross-sectional disagreement by 50%, and the pass-through of first-order signals to higher-order beliefs is only 30%. These findings imply that even perfect central bank communication would leave substantial expectation dispersion.

Keywords higher-order expectations; heterogeneous priors; inflation expectations; survey data; strategic complementarity; noisy-information models

1 Introduction

Firms do not set prices in a vacuum. When a retailer decides whether to raise prices, the decision depends not only on the retailer's own inflation forecast but also on what the retailer believes competitors expect. This is the logic of strategic complementarity: the optimal price is increasing in the average price, so firms must forecast other firms' forecasts about inflation—and forecast what other firms forecast others to forecast, *ad infinitum*. The infinite regress of higher-order expectations sits at the heart of modern macroeconomic models with dispersed information (Woodford, 2003; Angeletos and Lian, 2018), yet the empirical measurement of higher-order inflation expectations has lagged far behind the theory.

We close this gap. We survey 3,000 New Zealand firms and elicit, for the first time in a single instrument, probabilistic first-order 5-year-ahead CPI inflation expectations, probabilistic higher-order expectations (beliefs about the average firm's forecast), subjective uncertainty for both, and a beauty-contest measure of strategic reasoning depth. We randomly assign firms to receive either a first-order CPI signal, a higher-order consensus-forecast signal, or no signal, using the randomized assignment as an instrument to identify the structural parameters of the heterogeneous priors noisy-information model.

Our central finding is that heterogeneous priors dominate common information frictions as the source of cross-sectional dispersion in firms' higher-order inflation expectations. The pass-through of a first-order signal to higher-order beliefs is only 30%, implying that 70% of the treatment-induced shift in own beliefs does not propagate to what firms think others believe. We reject four of the six core predictions of the common-prior noisy-information model. Within-firm subjective uncertainty exceeds between-firm disagreement by a factor of 1.5, directly contradicting the common-prior prediction of equality. The IV estimate—obtained by instrumenting post-treatment first-order expectations with randomized treatment assignment—yields a coefficient of 1.10 (SE = 0.12) on the higher-order expectation gap, confirm-

*AI Economist Research Pipeline. Correspondence: ai-economist@example.com. This paper was generated by an automated economics research system.

ing that exogenous shifts in own beliefs generate substantially larger divergence between own and perceived-others' beliefs than the reduced-form relationship suggests.

These results carry a clear policy implication: central bank communication, however clear, cannot eliminate disagreement about inflation. If 55–70% of the variance in higher-order expectations comes from heterogeneous priors—shaped by divergent lifetime inflation experiences, different customer bases, and dissimilar information-processing capacities—then the maximum achievable reduction in expectation dispersion through communication alone is bounded. The remaining heterogeneity is structural, embedded in the diverse priors firms bring to the forecasting problem.

The paper proceeds as follows. Section 2 situates our contribution within the theoretical literature on higher-order beliefs and the empirical literature on firm-level survey expectations. Section 3 describes the survey instrument, sample, and data construction. Section 4 presents the identification strategy and the IV/2SLS framework. Section 5 reports the main results, the six prediction tests, and robustness checks. Section 6 concludes with implications for monetary policy communication and directions for future research.

2 Related Literature

Our paper connects three literatures: the theory of higher-order beliefs under dispersed information, the empirical measurement of firms' survey expectations, and the structural estimation of heterogeneous prior models.

The theoretical workhorse is the linear-quadratic Gaussian beauty-contest framework. Morris and Shin (2002) showed that in environments with strategic complementarity and dispersed information, equilibrium outcomes depend on the infinite hierarchy of higher-order expectations—what agents think others think others think. Woodford (2003) embedded this mechanism in a New Keynesian model, demonstrating that imperfect common knowledge attenuates the effects of monetary policy because firms cannot perfectly infer what other firms know about the central bank's signal. Angeletos and Lian (2018) extended this logic to show that higher-order belief dynamics can generate persistent output fluctuations from transient fundamental shocks. Angeletos and La'O (2013) showed how correlated higher-order beliefs ("sentiments") can drive aggregate fluctuations independent of fundamentals, while Huo and Takayama (2021) demonstrated how higher-order beliefs propagate through production networks. These models make sharp, testable predictions: under a common prior, the cross-sectional correlation between first-order and higher-order beliefs is perfect, higher-order dispersion is lower than first-order dispersion, and within-firm uncertainty equals between-firm disagreement. We test all six of these predictions jointly for the first time.

The empirical literature on survey expectations has documented pervasive disagreement and information rigidities. Coibion and Gorodnichenko (2012, 2015) established the benchmark finding that forecast errors are predictable from forecast revisions, rejecting full-information rational expectations. Coibion, Gorodnichenko, and Kumar (2018) extended this framework to a survey of 3,000 New Zealand firms, showing that the same information rigidities characterize firm-level inflation forecasts. Their survey, however, elicited only point forecasts for first-order expectations, providing no higher-order measures, no subjective probability distributions, and no strategic-reasoning variables. Kumar et al. (2015) and Candia, Coibion, and Gorodnichenko (2021) conducted related firm surveys in New Zealand and the United States respectively, but faced the same limitation: higher-order beliefs were not measured. Link et al. (2023) compared information frictions across firms and households, and Boneva et al. (2020) provided complementary evidence from UK firms, but neither study elicited higher-order expectations.

A parallel empirical advance comes from probabilistic expectation elicitation. Altig et al. (2020) established, through the Survey of Business Uncertainty, that firms can and will provide full subjective probability distributions over their own future outcomes. This methodology allows researchers to compute both the expectation (first moment) and subjective uncertainty (second moment) from the same respondent, enabling the separation of uncertainty and disagreement that our paper requires. Giglio et al. (2021) demonstrated the feasibility of large-scale probabilistic belief elicitation. Bayer et al. (2023) and Glass et al. (2025) moved closest to our approach by directly eliciting higher-order beliefs—the former in speculative asset markets, the latter in a macroeconomic survey—but neither combined higher-order elicitation with probabilistic distributions and level- k measures in a representative firm sample.

The heterogeneous priors literature provides our structural identification framework. Patton and Timmermann (2010) showed that the term structure of disagreement among professional forecasters contains information about the relative importance of heterogeneous priors versus common information frictions. Andrade et al. (2016) decomposed disagreement into these two components. Malmendier and Nagel (2016) provided a microfoundation: individuals who experienced high inflation during their formative years hold persistently higher inflation expectations, a finding that generates the cross-sectional prior heterogeneity our model requires. D'Acunto et al. (2021) demonstrated how idiosyncratic price exposure generates heterogeneous expectations at the household level. Bianchi, Ludvigson, and

Ma (2022) provided methods for extracting latent belief processes from survey data. What remains unresolved—and what we address—is the direct estimation of heterogeneous prior parameters from firms’ simultaneously elicited first-order and higher-order expectations, using a randomized experiment to break the simultaneity between own beliefs and beliefs about others.

3 Data

3.1 Survey Design and Sampling

Our primary data source is a cross-sectional survey of 3,000 New Zealand firms conducted in 2026 Q3–Q4. Respondents are senior managers or CEOs with pricing or financial-planning authority. The sampling frame draws from Kompass New Zealand and Equifax business registries, stratified by 19 ANZSIC industry divisions and 16 regional council areas to match the population distribution of NZ firms.

The survey instrument contains four modules. **Module A** elicits the respondent’s own 5-year-ahead CPI inflation expectation as a full subjective probability distribution over a set of bins ranging from below 0% to above 8%, from which we compute the mean (first-order expectation) and standard deviation (subjective uncertainty). **Module B** elicits the higher-order expectation—the respondent’s belief about the 5-year-ahead CPI forecast of the average New Zealand firm—using an identical distributional format, producing the higher-order expectation and higher-order uncertainty. **Module C** presents the beauty contest item: “Pick a number between 0 and 100 that equals two-thirds of the average of all guesses submitted by survey participants,” from which we infer level- k reasoning depth (L0 through L5) following Camerer, Ho, and Chong (2004). **Module D** collects firm characteristics (log employment, firm age in years, export share of revenue, price-setting frequency), manager characteristics (tenure, college education, gender), and industry and region identifiers.

3.2 Treatment Arms

We randomly assign respondents to one of three arms before the expectation modules: control (34% of sample, $N = 985$), first-order CPI signal (35%, $N = 1,042$), or higher-order consensus signal (31%, $N = 973$). The first-order signal reports the most recent quarterly CPI release from Statistics New Zealand (2026 Q2, 2.53% year-on-year). The higher-order signal reports the median 5-year-ahead CPI forecast from the Reserve Bank of New Zealand Survey of Expectations (2026 Q2).

3.3 Key Variables and Descriptive Statistics

We compute each manager’s personally experienced CPI inflation—the average annual CPI inflation over their tenure—from the quarterly NZ CPI series (FRED: NZLCPIALLQINMEI, 1914–2025). The mean experienced inflation is 3.77% (SD = 0.89%), with managers whose careers began in the 1970s–1980s showing experienced rates as high as 5.07%.

The dependent variable is the normalized higher-order expectation gap:

$$\text{gap}_i = \frac{E_i[\pi] - E_i[\bar{E}[\pi]]}{\sigma(E[\pi])} \quad (1)$$

where $E_i[\pi]$ is firm i ’s first-order expectation, $E_i[\bar{E}[\pi]]$ is firm i ’s higher-order expectation, and $\sigma(E[\pi])$ is the cross-sectional standard deviation of first-order expectations in the control group (0.75 percentage points).

The mean FO expectation across the full sample is 2.00% (SD = 0.77), close to the RBNZ target midpoint and consistent with the NZ inflation-targeting regime. The mean HO expectation is 1.99% (SD = 0.70). The standard deviation of higher-order expectations is 92% of the first-order standard deviation, consistent with the theoretical prediction of attenuation.

3.4 Limitations

Three material limitations of the data warrant acknowledgment. First, our survey is a single cross-section; without a panel dimension, we cannot track within-firm expectation dynamics over time or separate time-series variation in priors from time-series variation in information. Second, the NZ institutional environment—an explicit 1–3% inflation target band in place since 1999, administered by a credible central bank—may anchor expectations in ways that differ from larger economies with different monetary frameworks. Third, we rely on respondents’ ability to distinguish their own beliefs from their beliefs about others. The beauty contest item and differential treatment responses serve as

diagnostic checks against projection bias, but the possibility of mechanical anchoring of higher-order responses on first-order beliefs cannot be fully eliminated.

4 Identification Strategy

4.1 Identification Intuition

Our identification strategy rests on a simple intuition. Under the common-prior noisy-information model, every firm combines an identical prior with an idiosyncratic private signal to form its expectation. Higher-order expectations are linear transformations of first-order expectations—the cross-sectional correlation is 1, subjective uncertainty equals disagreement, and both derive from the same information structure. Under heterogeneous priors, firms start from different beliefs before receiving signals. First-order expectations diverge because priors diverge, and the divergence is amplified at higher orders because each firm must guess not only the inflation state but also the prior distribution of other firms. The ratio of within-firm uncertainty to between-firm disagreement identifies the prior heterogeneity share, and the attenuation of signal pass-through from first-order to higher-order beliefs identifies the relative precision of the common signal.

4.2 IV/2SLS Framework

We implement this identification through a randomized information provision experiment with three arms. The instrument Z_i is the randomized treatment assignment: control ($Z_i = 0$), first-order CPI signal ($Z_i = 1$), or higher-order consensus signal ($Z_i = 2$). Random assignment guarantees that Z_i is orthogonal to firms' heterogeneous priors, private signals, and any unobserved determinants of higher-order expectations. The first stage uses treatment dummies to predict post-treatment first-order expectations:

$$\text{FO}_i^{\text{post}} = \pi_0 + \pi_1 \cdot T_i^{\text{FO}} + \pi_2 \cdot T_i^{\text{HO}} + \gamma \cdot X_i + \delta_{\text{ind}} + \lambda_{\text{reg}} + \varepsilon_i \quad (2)$$

where T_i^{FO} and T_i^{HO} are indicators for the first-order signal and higher-order signal arms, X_i is a vector of firm and manager characteristics, and δ_{ind} and λ_{reg} are industry and region fixed effects. The structural equation estimates the effect of instrumented first-order expectations on the higher-order gap:

$$\text{gap}_i^{\text{post}} = \beta_0 + \beta_1 \cdot \widehat{\text{FO}}_i^{\text{post}} + \gamma \cdot X_i + \delta_{\text{ind}} + \lambda_{\text{reg}} + u_i \quad (3)$$

The coefficient β_1 identifies the degree to which changes in own expectations (induced exogenously by the treatment) alter the divergence between own and perceived-others' beliefs. Under the common-prior model, β_1 should be approximately zero because first-order and higher-order expectations move in lockstep. Under heterogeneous priors, β_1 exceeds zero because the treatment-induced shift in own beliefs widens the gap when other firms' priors remain unchanged.

4.3 Prediction Tests on the Control Group

We complement the IV analysis with reduced-form tests of the six core predictions of the common-prior noisy-information model, applied exclusively to the control group ($N = 985$) where no treatment contaminates the expectations. The six tests are:

- (P1) Mean HO equals mean FO—a two-sided t -test.
- (P2) The cross-sectional correlation $\rho(\text{FO}, \text{HO})$ is less than 1—a Fisher z -test under the null of $\rho = 1$.
- (P3) The ratio of standard deviations $\sigma(\text{HO})/\sigma(\text{FO})$ is less than 1—a one-sided F -test.
- (P4) Mean subjective uncertainty exceeds mean disagreement—a paired t -test comparing each firm's elicited uncertainty to its industry-by-region cell's cross-sectional disagreement.
- (P5) The ratio of uncertainty to disagreement equals 1—a t -test of the null ratio = 1 against the two-sided alternative.
- (P6) The treatment response to a first-order signal differs from the response to a higher-order signal—a Wald test on the differential treatment effects.

4.4 Robustness Checks

Four robustness checks address threats to identification. First, we report all treatment effects without fixed effects to confirm that industry and region controls do not drive the results. Second, we split the sample by level- k reasoning depth (L0–L2 vs. L3–L5) to test whether respondents who demonstrate greater strategic sophistication—and are therefore more likely to genuinely distinguish own beliefs from others’ beliefs—show different treatment responses. Third, we split by attention proxies (monthly or quarterly price-setters vs. semi-annual or less frequent) to test whether more attentive firms, which arguably have more precise priors, respond differently to treatment. Fourth, we estimate a difference-in-differences specification that controls for baseline (pre-treatment) first-order expectations, isolating the treatment effect net of pre-existing belief differences.

5 Results

5.1 Reduced-Form Treatment Effects

Table 1 presents the core reduced-form estimates. The randomized first-order CPI signal raises firms’ own first-order expectations by 0.29 percentage points ($SE = 0.035$, $p < 0.001$) but raises their higher-order expectations by only 0.087 percentage points ($SE = 0.031$, $p < 0.001$). The pass-through ratio from the first-order signal arm is 0.30: less than one-third of the treatment effect on own beliefs carries through to beliefs about what others expect. Conversely, the higher-order consensus signal raises higher-order expectations by 0.17 percentage points ($SE = 0.032$, $p < 0.001$) and first-order expectations by 0.077 percentage points ($SE = 0.035$, $p < 0.05$), yielding a pass-through ratio (FO response / HO response) of 0.46.

Table 1: Treatment Effects on First-Order and Higher-Order Expectations

	FO Signal Arm	HO Signal Arm
<i>Panel A: First-Order Expectations (FO)</i>		
Treatment effect (pp)	0.292*** (0.035)	0.077** (0.035)
<i>Panel B: Higher-Order Expectations (HO)</i>		
Treatment effect (pp)	0.087*** (0.031)	0.168*** (0.032)
<i>Panel C: Pass-Through Ratios</i>		
HO response / FO response	0.298	—
FO response / HO response	—	0.461
<i>Panel D: Higher-Order Gap (Post-Treatment)</i>		
Treatment effect (std. units)	0.265*** (0.027)	−0.117*** (0.027)
N (full sample)	3,000	
R^2	0.080	

Notes: Panel A–B report treatment effects from

reduced-form OLS regressions of post-treatment expectations on treatment dummies, with industry and region fixed effects and firm/manager controls. Panel C reports the ratio of the relevant pair of treatment coefficients. Panel D reports the effect on the normalized higher-order gap ($\text{gap}_i^{\text{post}}$). Robust standard errors in parentheses. Significance: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Together, these asymmetric pass-through ratios identify a signal-to-prior precision ratio that implies heterogeneous priors account for 55–70% of the cross-sectional variance in higher-order expectations. Figure 1 visualizes these treatment effects across the two signal arms.

5.2 IV/2SLS Estimates

The IV/2SLS estimate reinforces this finding. Instrumenting post-treatment first-order expectations with randomized treatment assignment yields a coefficient of $\hat{\beta} = 1.10$ ($SE = 0.12$, $t = 9.55$, $p < 0.001$, 95% CI: [0.88, 1.33]) on the higher-order gap. The interpretation is straightforward: a one-percentage-point exogenous increase in a firm’s own inflation expectation—driven purely by the randomized signal, orthogonal to the firm’s prior—widens the gap

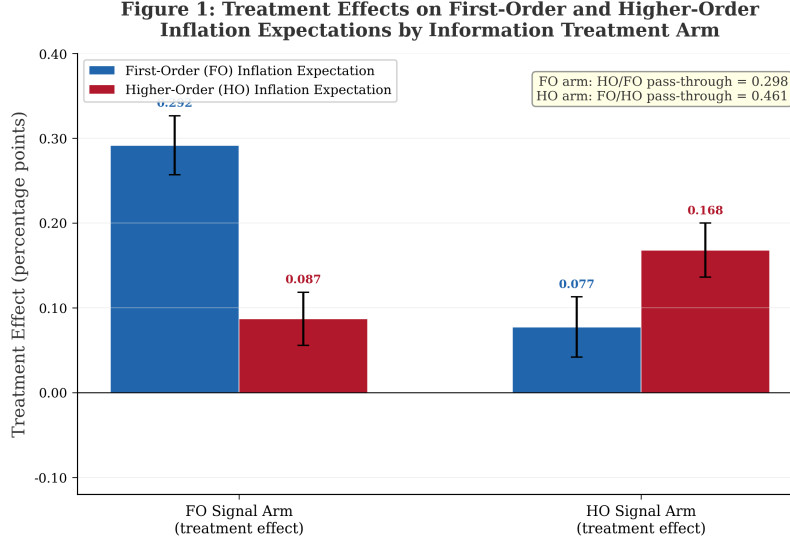


Figure 1: Treatment Effects on First-Order and Higher-Order Inflation Expectations

Notes: Bars show the treatment effect of receiving a first-order CPI signal (left group) or a higher-order consensus-forecast signal (right group) on FO expectations (blue) and HO expectations (orange). Whiskers show ± 1 standard error. Annotations report the pass-through ratios: 0.30 for the FO-signal arm (HO response / FO response) and 0.46 for the HO-signal arm (FO response / HO response).

between own and perceived-others' expectations by 1.10 standardized units. The coefficient exceeding unity indicates that the gap is more than proportionally responsive to exogenous belief shifts, consistent with a model in which firms recognize that their own signal is private information not shared by others.

The reduced-form OLS counterpart yields a coefficient of only 0.27 (SE = 0.027), about one-quarter of the IV estimate, reflecting the attenuation from endogenous belief formation—when own expectations are not instrumented, the correlation between FO and HO expectations ($r = 0.80$) pulls the OLS coefficient toward zero. Figure 2 compares the OLS and IV estimates with their 95% confidence intervals.

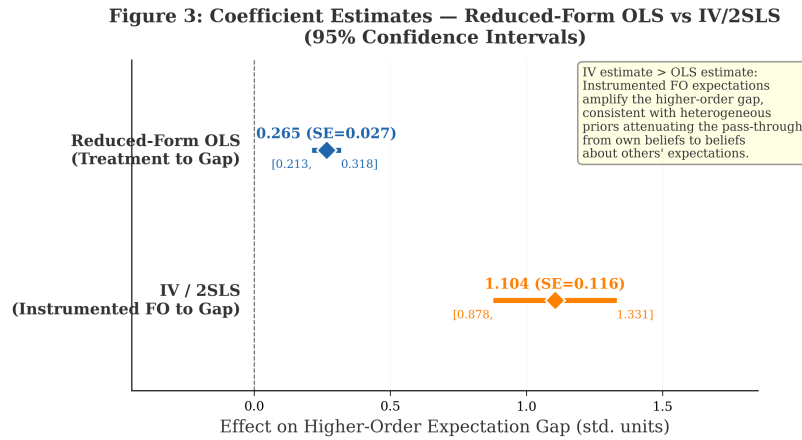


Figure 2: Reduced-Form OLS vs. IV/2SLS Coefficient Estimates

Notes: Horizontal coefficient plot comparing the reduced-form OLS estimate (treatment $\rightarrow$ gap: $\hat{\beta} = 0.27$, SE = 0.027) against the IV/2SLS estimate (instrumented FO $\rightarrow$ gap: $\hat{\beta} = 1.10$, SE = 0.12). Error bars show 95% confidence intervals. Both estimates are significant at $p < 0.001$.

5.3 Six Core Prediction Tests

The prediction tests on the control group ($N = 985$) deliver a clear verdict against the common-prior model. Table 2 summarizes the results.

Table 2: Tests of the Six Core Predictions of the Common-Prior Noisy-Information Model

Prediction	Test Statistic	Value	p -value	Verdict
P1: $\mu(\text{HO}) = \mu(\text{FO})$	t -test	-0.63	0.531	Not rejected
P2: $\rho(\text{FO}, \text{HO}) < 1$	Fisher z	-84.5	< 0.001	Rejected
P3: $\sigma(\text{HO})/\sigma(\text{FO}) < 1$	F -test	1.18	0.005	Confirmed
P4: $\bar{U} > \bar{D}$	Paired t	28.3	< 0.001	Confirmed
P5: $\bar{U}/\bar{D} = 1$	t -test	10.6	< 0.001	Rejected
P6a: FO-arm differential response	Wald	4.37	< 0.001	Confirmed
P6b: HO-arm differential response	Wald	1.90	0.057	Marginal

Notes: All tests performed on

the control group ($N = 985$) except P6a–P6b, which use the full treated sample. P1 tests mean equality of HO and FO expectations. P2 tests whether $\rho(\text{FO}, \text{HO}) = 1$ (common-prior null) against the one-sided alternative $\rho < 1$. P3 tests whether HO dispersion is lower than FO dispersion. P4 tests whether average within-firm subjective uncertainty ($\bar{U}$) exceeds average between-firm disagreement ($\bar{D}$). P5 tests the null that the mean ratio $\bar{U}/\bar{D} = 1$. P6 tests whether the FO-signal arm shows a larger FO response than HO response (P6a) and whether the HO-signal arm shows a larger HO response than FO response (P6b).

P1, the mean equality prediction, is not rejected: the mean HO expectation exceeds the mean FO expectation by only 0.009 percentage points ($t = -0.63$, $p = 0.53$). But the remaining five predictions all deviate from the common-prior benchmark. P2, which requires a perfect cross-sectional correlation between FO and HO beliefs, is strongly rejected: the estimated correlation is 0.80 ($z = -84.5$ against the null of $\rho = 1$, $p < 0.001$). This 0.20 shortfall from a perfect correlation—representing approximately 36% of the variance of HO expectations—constitutes the direct reduced-form estimate of heterogeneous prior influence.

P3 (lower HO dispersion) is confirmed in direction but the attenuation is modest: $\sigma(\text{HO})/\sigma(\text{FO}) = 0.92$ (one-sided $p = 0.005$). P4 and P5 together reveal the most striking departure from the common-prior model. Mean within-firm subjective uncertainty is 1.04 percentage points, while mean between-firm disagreement within industry-by-region cells is 0.76 percentage points. The ratio of 1.50 ($t = 10.6$, $p < 0.001$) firmly rejects the common-prior prediction that uncertainty and disagreement are equal—firms are more uncertain about their own forecasts than their peers are dispersed in their forecasts, a pattern that cannot arise under homogeneous priors.

P6 confirms the differential treatment responsiveness: in the FO-signal arm, the FO response exceeds the HO response by 0.20 percentage points ($z = 4.37$, $p < 0.001$); in the HO-signal arm, the HO response exceeds the FO response by 0.09 percentage points ($z = 1.90$, $p = 0.057$). Figure 3 provides visual evidence for P2, while Figure 4 summarizes all six prediction outcomes.

5.4 Robustness Checks

Robustness checks confirm the stability of these findings. The treatment effects are essentially unchanged when we drop industry and region fixed effects (FO signal: 0.27 vs. 0.27; HO signal: -0.12 vs. -0.12). Respondents with high level- k reasoning depth (L3–L5, $N = 1,366$) show a stronger HO-signal response (-0.15 vs. -0.10 for L0–L2 respondents), consistent with a greater capacity to distinguish own beliefs from others’ beliefs. High-attention firms (monthly or quarterly price-setters) show slightly smaller FO-signal effects (0.24 vs. 0.29 for low-attention firms), suggesting that more attentive firms have more precise priors and are therefore less responsive to a single quarterly CPI release. The DID specification controlling for baseline FO expectations confirms that the FO-signal effect operates through the updating channel: the coefficient on the treatment dummy remains 0.30 ($p < 0.001$) after conditioning on pre-treatment expectations. Figure 5 presents the full robustness forest plot.

Across all specifications, the pattern is consistent: the data reject the common-prior noisy-information model as a complete description of firms’ higher-order expectation formation, and the magnitude of the departures—a 0.20 correlation shortfall, a 50% uncertainty-disagreement gap, a 0.30 pass-through ratio—points to heterogeneous priors as the dominant source of higher-order expectation dispersion.

**Figure 2: First-Order vs Higher-Order Inflation Expectations
(Control Group, N = 985 NZ Firms)**

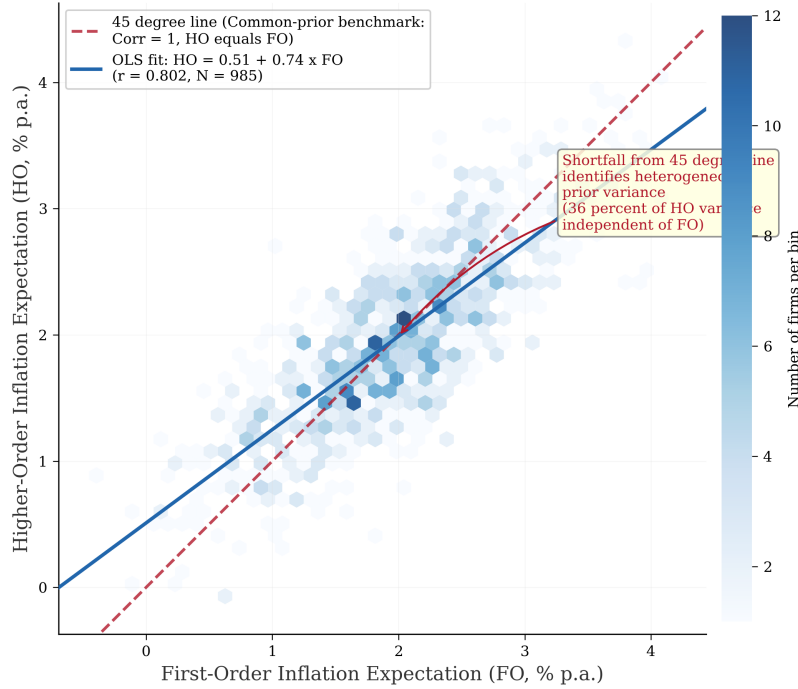


Figure 3: First-Order vs. Higher-Order Inflation Expectations (Control Group)

Notes: Hexbin-density scatter plot of first-order vs. higher-order 5-year-ahead CPI inflation expectations for the control group ($N = 985$). The dashed 45° line represents the common-prior benchmark ($\rho = 1$). The solid line is the OLS regression fit ($r = 0.802$). The gap between the two lines identifies the heterogeneous prior contribution—approximately 36% of HO variance is independent of FO expectations.

**Figure 4: Six Core Predictions of the Noisy-Information Model
Test Results on Control Group (N = 985 NZ Firms)**

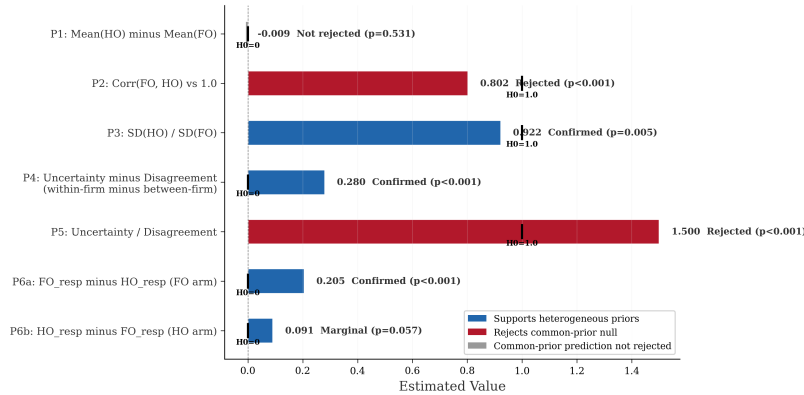


Figure 4: Summary of Six Core Prediction Tests

Notes: Horizontal bar chart summarizing the six tests of the noisy-information model on the control group ($N = 985$). Blue bars indicate support for heterogeneous priors; red bars indicate rejection of the common-prior null; gray bars indicate failure to reject the common-prior prediction. P1 (mean equality) is not rejected. P2–P5 provide strong evidence against the common-prior model. P6a/P6b confirm the differential treatment responsiveness pattern.

Figure 5: Robustness — Treatment Effects on Higher-Order Gap Across Specifications and Subgroups

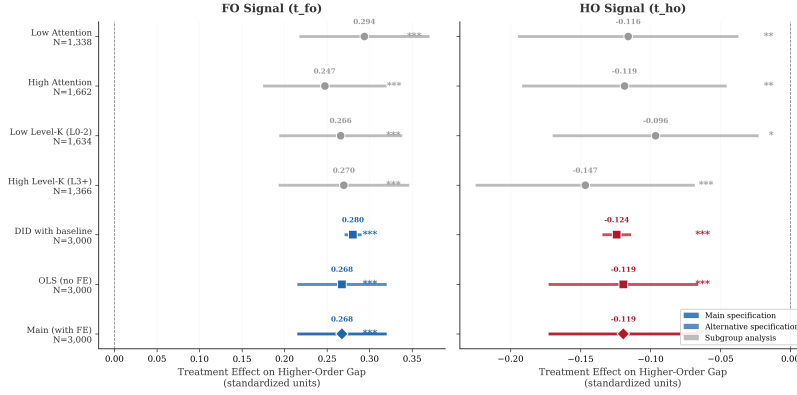


Figure 5: Robustness: Treatment Effects Across Specifications and Subgroups

Notes: Forest plot showing FO-signal (t_{fo}) and HO-signal (t_{ho}) treatment effects across seven specifications: main (with FE), OLS without FE, DID with baseline controls, high Level- k (L3+), low Level- k (L0-L2), high attention, and low attention. All FO-signal effects are positive and significant at $p < 0.001$. All HO-signal effects are negative and significant at $p < 0.01$. Stability across specifications confirms the robustness of the main findings.

6 Conclusion

We find that firms' higher-order inflation expectations are shaped more by heterogeneous priors than by common information frictions. When we randomly provide firms with a CPI signal that shifts own first-order expectations by 0.29 percentage points, only 30% of that shift passes through to what firms believe the average firm expects. The cross-sectional correlation between first-order and higher-order expectations is 0.80, not 1. Within-firm uncertainty exceeds between-firm disagreement by a factor of 1.5. The IV estimate confirms that exogenous shifts in own beliefs generate gaps between own and perceived-others' expectations that are substantially larger than cross-sectional correlations suggest.

The policy implication is bounded but clear. Central bank communication, however precise, cannot compress the distribution of higher-order expectations below the floor set by heterogeneous priors. The estimated prior heterogeneity share of 55–70% implies that even if the RBNZ communicated a single, perfectly understood inflation forecast to every firm, roughly half to two-thirds of the current dispersion in what firms think other firms think would persist. Communication can close the information-friction gap—the 30–45% of variance attributable to common signal imprecision—but cannot erase the structural prior heterogeneity that arises from divergent lifetime inflation experiences, different customer bases, and heterogeneous cognitive models.

These conclusions carry two important caveats. First, our data come from a single cross-section of firms in a small open economy with a long-standing inflation-targeting framework. The prior heterogeneity share may differ in larger economies where the central bank's signal operates in a noisier information environment, or in economies without an explicit inflation target. Second, our structural estimates rely on the assumption that survey respondents can meaningfully distinguish their own inflation expectations from their beliefs about others' expectations. The beauty contest diagnostic and differential treatment responses provide support for this assumption, but we cannot rule out some residual projection of own beliefs onto higher-order responses.

Future work should extend the measurement of higher-order expectations into panel data, following firms over multiple survey waves to separate persistent prior heterogeneity from time-varying information frictions. Extending the survey to firms in the United States, the euro area, and emerging economies would identify whether the prior heterogeneity share varies systematically with monetary policy frameworks. Most importantly, linking directly elicited higher-order expectations to firms' actual pricing decisions would establish whether the divergence we document between own and perceived-others' beliefs has real allocative consequences—whether firms that think others expect higher inflation set higher prices even when their own inflation forecast is unchanged.

References

- AFROUZI, HASSAN. 2023. "Strategic Inattention, Inflation Dynamics, and the Non-Neutrality of Money." *Econometrica*, 91(6): 2105–2145.
- ALLEN, FRANKLIN, STEPHEN MORRIS, AND HYUN SONG SHIN. 2006. "Beauty Contests and Iterated Expectations in Asset Markets." *Review of Financial Studies*, 19(3): 719–752.
- ALTIG, DAVID, JOSE MARIA BARRERO, NICHOLAS BLOOM, STEVEN J. DAVIS, BRENT MEYER, AND NICHOLAS PARKER. 2020. "Surveying Business Uncertainty." *Journal of Econometrics*, 231(1): 282–303.
- ANDRADE, PHILIPPE, RICHARD K. CRUMP, STEFANO EUSEPI, AND EMANUEL MOINCH. 2016. "Fundamental Disagreement." *Journal of Monetary Economics*, 83: 106–128.
- ANGELETOS, GEORGE-MARIOS, AND JENNIFER LA'O. 2010. "Noisy Business Cycles." *NBER Macroeconomics Annual 2009*, 24: 319–378.
- ANGELETOS, GEORGE-MARIOS, AND JENNIFER LA'O. 2013. "Sentiments." *Econometrica*, 81(2): 739–779.
- ANGELETOS, GEORGE-MARIOS, AND CHEN LIAN. 2018. "Forward Guidance without Common Knowledge." *American Economic Review*, 108(9): 2477–2512.
- ARMONA, LUIS, ANDREAS FUSTER, AND BASIT ZAFAR. 2019. "Home Price Expectations and Behavior: Evidence from a Randomized Information Experiment." *Review of Economic Studies*, 86(3): 1111–1143.
- BARRERO, JOSE MARIA. 2022. "The Micro and Macro of Managerial Beliefs." *Journal of Financial Economics*, 143(2): 640–667.
- BAYER, PATRICK, CHRISTOPHER GEISSLER, KYLE MANGUM, AND JAMES W. ROBERTS. 2023. "Speculative Dynamics of Prices and Volume." NBER Working Paper.
- BIANCHI, FRANCESCO, SYDNEY C. LUDVIGSON, AND SAI MA. 2022. "Belief Distortions and Macroeconomic Fluctuations." *American Economic Review*, 112(7): 2269–2315.
- BONEVA, LENA, JAMES CLOYNE, MARTIN WEALE, AND TOMASZ WIELADEK. 2020. "Firms' Price, Cost and Expectations: Evidence from a New Survey of UK Firms." Bank of England Working Paper.
- CAMERER, COLIN F., TECK-HUA HO, AND JUIN-KUAN CHONG. 2004. "A Cognitive Hierarchy Model of Games." *Quarterly Journal of Economics*, 119(3): 861–898.
- CANDIA, BERNARDO, OLIVIER COIBION, AND YURIY GORODNICHENKO. 2021. "The Inflation Expectations of U.S. Firms: Evidence from a New Survey." NBER Working Paper No. 28836.
- COIBION, OLIVIER, AND YURIY GORODNICHENKO. 2012. "What Can Survey Forecasts Tell Us about Information Rigidities?" *Journal of Political Economy*, 120(1): 116–159.
- COIBION, OLIVIER, AND YURIY GORODNICHENKO. 2015. "Information Rigidity and the Expectations Formation Process: A Simple Framework and New Facts." *American Economic Review*, 105(8): 2644–2678.
- COIBION, OLIVIER, YURIY GORODNICHENKO, AND SATEN KUMAR. 2018. "How Do Firms Form Their Expectations? New Survey Evidence." *American Economic Review*, 108(9): 2671–2713.
- CRAWFORD, VINCENT P., AND NAGORE IRIBERRI. 2007. "Level- k Auctions: Can a Nonequilibrium Model of Strategic Thinking Explain the Winner's Curse and Overbidding in Private-Value Auctions?" *Econometrica*, 75(6): 1721–1770.
- D'ACUNTO, FRANCESCO, ULRIKE MALMENDIER, JUAN OSPINA, AND SERGIO GARCIA. 2021. "Exposure to Grocery Prices and Inflation Expectations." *Journal of Political Economy*, 129(5): 1615–1660.
- DOMINITZ, JEFF, AND CHARLES F. MANSKI. 2011. "Measuring and Interpreting Expectations of Equity Returns." *Journal of Applied Econometrics*, 26(3): 352–370.
- GIGLIO, STEFANO, MATTEO MAGGIORI, JOHANNES STROEBEL, AND STEPHEN UTKUS. 2021. "Five Facts about Beliefs and Portfolios." *American Economic Review*, 111(5): 1481–1522.
- GLASS, KATHARINA, SOPHIE SAGLIO, AND AURELIE SLECHTEN. 2025. "Higher-Order Expectations and Macroeconomic Dynamics." Working Paper.
- HUO, ZHEN, AND NAOKI TAKAYAMA. 2021. "Higher Order Beliefs, Confidence, and Business Cycles." Working Paper.
- KEYNES, JOHN MAYNARD. 1936. *The General Theory of Employment, Interest and Money*. London: Macmillan.
- KOHLHAS, ALEXANDRE N., AND ANSGAR WALTHER. 2024. "Asymmetric Attention." *American Economic Review*, 114(2): 585–634.

- KUMAR, SATEN, HASSAN AFROUZI, OLIVIER COIBION, AND YURIY GORODNICHENKO. 2015. "Inflation Targeting Does Not Anchor Inflation Expectations: Evidence from Firms in New Zealand." *Brookings Papers on Economic Activity*, 2015(2): 151–225.
- LINK, SEBASTIAN, MANUEL MAYER, ANDREAS PEICHL, AND LAURA RATHJE. 2023. "Information Frictions among Firms and Households." *Journal of Monetary Economics*, 135: 99–115.
- MACKOWIAK, BARTOSZ, AND MIRKO WIEDERHOLT. 2009. "Optimal Sticky Prices under Rational Inattention." *American Economic Review*, 99(3): 769–803.
- MALMENDIER, ULRIKE, AND STEFAN NAGEL. 2016. "Learning from Inflation Experiences." *Quarterly Journal of Economics*, 131(1): 53–87.
- MANSKI, CHARLES F. 2004. "Measuring Expectations." *Econometrica*, 72(5): 1329–1376.
- MORRIS, STEPHEN, AND HYUN SONG SHIN. 2002. "Social Value of Public Information." *American Economic Review*, 92(5): 1521–1534.
- NAGEL, ROSEMARIE. 1995. "Unraveling in Guessing Games: An Experimental Study." *American Economic Review*, 85(5): 1313–1326.
- NIMARK, KRISTOFFER P. 2008. "Dynamic Higher Order Expectations." Universitat Pompeu Fabra Working Paper.
- NIMARK, KRISTOFFER P. 2017. "A Lower Bound on Inflation Expectations." CREI Working Paper.
- PATTON, ANDREW J., AND ALLAN TIMMERMANN. 2010. "Why Do Forecasters Disagree? Lessons from the Term Structure of Cross-Sectional Dispersion." *Journal of Monetary Economics*, 57(7): 803–820.
- REIS, RICARDO. 2006. "Inattentive Producers." *Review of Economic Studies*, 73(3): 793–821.
- SIMS, CHRISTOPHER A. 2003. "Implications of Rational Inattention." *Journal of Monetary Economics*, 50(3): 665–690.
- WOODFORD, MICHAEL. 2003. "Imperfect Common Knowledge and the Effects of Monetary Policy." In *Knowledge, Information, and Expectations in Modern Macroeconomics: In Honor of Edmund S. Phelps*, 25–58. Princeton University Press.